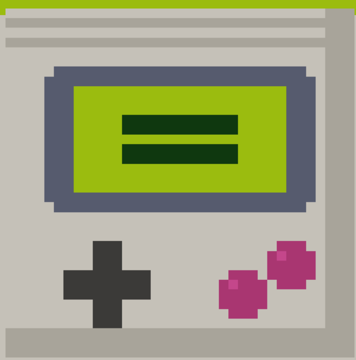
CALC BOY

THE MANUAL



Nintendo-style calculator PWA

Version 3.1.0 · GPL-3.0 · schrotty74.github.io/CalcBoy

CONTENTS

1. GETTING STARTED

Open https://schrotty74.github.io/CalcBoy/ in your browser. On iPhone: open it in Safari, tap Share → Add to Home Screen – CALC BOY then launches as a full-screen app. After the first load everything works completely offline.

A short boot animation plays on startup. Numbers use the German format: comma as decimal separator, dots as thousands separators. Up to 12 digits can be entered; long results shrink automatically. Impossible operations (e.g. division by 0) show ERROR – the next digit starts fresh.

2. OPERATING CONCEPT

The app has three areas: the pill buttons on top (THEME, GAME, SND), the LCD and the keypad. The keypad has seven pages:

BASIC → EXT → CONV → FIN → PRG → PLOT → FORM → BASIC

EXT	EXT (bottom left on the BASIC page) opens the first extra page.
MEHR	MEHR (on every extra page) cycles to the next page.
BASIC	BASIC (on every extra page) returns to the standard keypad.
=	= works on every page.

3. THE THREE TOP BUTTONS

THEME	THEME – switches to the next console look. Six themes are available: Game Boy (default), GB Color, NES, Super NES, Switch and Famicom; a seventh (Virtual Boy) is unlockable (chapter 14). The name flashes in the LCD, switching plays a cartridge-swap animation, and each theme has its own key sound. Your choice is saved.
GAME	GAME – opens the game menu (chapter 12). During a running game the button quits the game.
SND ON	SND – toggles the 8-bit key sounds on (SND ON) or off (SND OFF, button dims). Saved.

4. THE DISPLAY (LCD)

The LCD has two lines: the status line on top (running calculation, messages, game state) and the result line below. A small M in the top-left corner means the memory (M+/M−) holds a non-zero value.

Tapping the display opens the history:

• Shows the last 10 calculations with their results.
• Tap an entry to bring its result back into the display.
• » TEILEN / EXPORT exports the history as text via the share sheet.
• » PNG TEILEN renders the history as an image and shares/downloads it as a PNG file.
• Below the entries you see Σ (sum of all results) and Ø (their average).
• Tapping again (or any key) closes the history.

Long-pressing (~0.5 s) the display copies the current result to the clipboard (KOPIERT appears).

5. BASIC PAGE

0–9 – digit entry, maximum 12 digits. Example: type 7 4 2 → display 742.

, – decimal comma; only one comma per number. Example: 3 , 1 4 → display 3.14.

+ – addition. Example: 2 + 3 = → 5.

− – subtraction. Example: 9 − 3 = → 6.

× – multiplication. Example: 6 × 7 = → 42.

÷ – division. Example: 20 ÷ 4 = → 5. Division by 0 → ERROR.

Chained calculations evaluate left to right: 2 + 3 × 4 = gives 20 (first 2+3, then ×4) – like a classic desk calculator, without operator precedence.

= – evaluates and writes the calculation to the history. (In RPN mode: pushes the number onto the stack, chapter 9.)

AC – clears the entry and the running calculation. Also exits games and menus; in RPN mode AC additionally clears the stack.

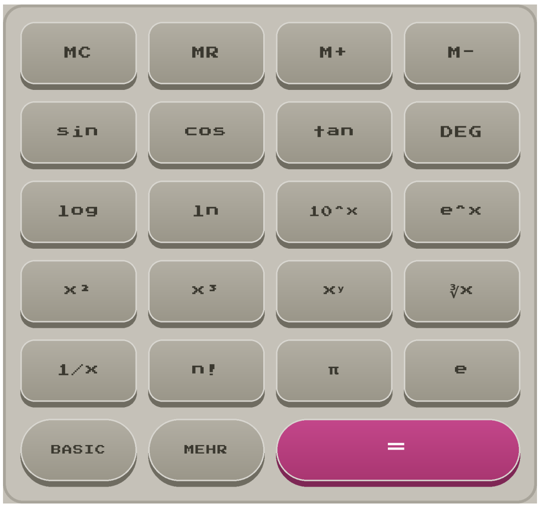
± – toggles the sign of the displayed number. Example: 5 → ± → −5.

% – smart percent key with three cases: After + or −: percent of the first operand, 100 + 10 % = 110; 200 − 25 % = 150. After × or ÷: converts to the factor, 50 × 10 % = 5. Standalone: divides by 100, 10 % = 0.1.

√ – square root of the displayed number. Negative input → ERROR.

EXT – switches to the extra pages.

6. EXT PAGE (SCIENTIFIC)



Memory

MC – clears the memory; the M indicator in the LCD disappears.

MR – recalls the memory value into the display. Example: 50 → M+, then AC, then MR → 50.

M+ – adds the displayed number to memory. Example: 50 → M+ (memory 50), later 25 → M+ → memory 75.

M− – subtracts the displayed number from memory. Example: memory 75, then 15 → M− → memory 60. Memory survives restarts; while it is ≠ 0 the LCD shows an M.

Trigonometry

sin – sine of the displayed number. Example (DEG): 30 → sin → 0.5.

cos – cosine. Example (DEG): 60 → cos → 0.5.

tan – tangent. Example (DEG): 45 → tan → 1.

DEG/RAD – toggles the angle mode: DEG = degrees, RAD = radians. Example: 180 → sin gives 0 in DEG, −0.80 in RAD. The button shows the active mode; the setting is saved and applies to all trigonometry keys.

Logarithms & exponentials

log – base-10 logarithm. Example: 1000 → log → 3.

ln – natural logarithm (base e). Example: 2.7182818285 → ln → 1.

10^x – power of ten. Example: 3 → 10^x → 1000.

e^x – e to the power of x. Example: 1 → e^x → 2.7182818285.

Powers & roots

x^2 – square of the displayed number. Example: 12 → x^2 → 144.

x^3 – third power. Example: 5 → x^3 → 125.

x^y – arbitrary power as a two-step operation: enter the base → press x^y → enter the exponent → =. Example: 2 x^y 10 = → 1,024.

$\sqrt[3]{x}$ – cube root. Example: 27 → $\sqrt[3]{x}$ → 3.

Miscellaneous

1/x – reciprocal. Example: 8 → 1/x → 0.125.

n! – factorial (1·2·…·n), integers 0–170 only, otherwise ERROR. Example: 6 → n! → 720.

π – puts the circle constant 3.1415926536 into the display, e.g. for circle calculations.

e – puts Euler's number 2.7182818285 into the display.

7. CONV PAGE (UNITS)

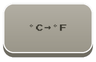
Principle: enter a value, tap a conversion key – the result replaces the display, and the status line names the conversion. Every conversion can be reversed with its counterpart key.

 km→mi – kilometers to miles (×0.621371). Example: 100 → 62.14.

 mi→km – miles to kilometers (×1.609344). Example: 10 → 16.09.

 m→ft – meters to feet (×3.28084). Example: 2 → 6.56.

 ft→m – feet to meters (×0.3048). Example: 10 → 3.05.

 °C→°F – Celsius to Fahrenheit (×9/5+32). Example: 25 → 77.

 °F→°C – Fahrenheit to Celsius ((−32)×5/9). Example: 98.6 → 37.

 kg→lb – kilograms to pounds (×2.204623). Example: 75 → 165.35.

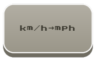
 lb→kg – pounds to kilograms (×0.453592). Example: 10 → 4.54.

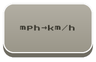
 cm→in – centimeters to inches (÷2.54). Example: 180 → 70.87.

 in→cm – inches to centimeters (×2.54). Example: 27 → 68.58.

 l→gal – liters to US gallons (×0.264172). Example: 50 → 13.21.

 gal→l – US gallons to liters (×3.785412). Example: 10 → 37.85.

 km/h→mph – speed (×0.621371). Example: 130 → 80.78.

 mph→km/h – speed (×1.609344). Example: 65 → 104.61.

 h→min – hours to minutes (×60). Example: 1.5 → 90.

 min→h – minutes to hours (÷60). Example: 45 → 0.75.

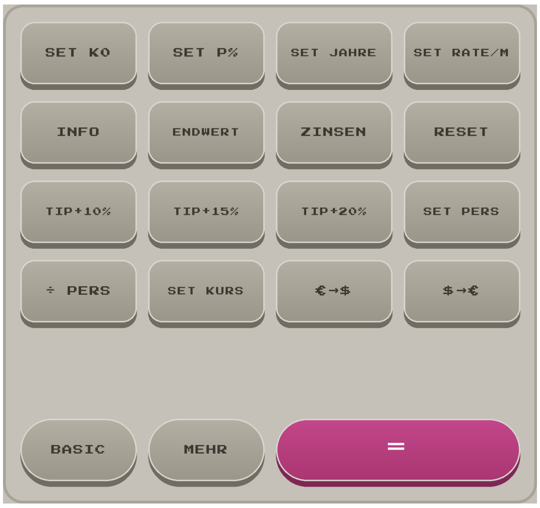
 MW+19 – net→gross with 19 % German VAT (×1.19). Example: 100 → 119.

 MW−19 – gross→net with 19 % German VAT (÷1.19, not −19 %!). Example: 119 → 100.

 MW+7 – net→gross with the reduced 7 % rate (×1.07). Example: 100 → 107.

 MW−7 – gross→net with the reduced 7 % rate (÷1.07). Example: 107 → 100.

8. FIN PAGE (FINANCE)



This page works with stored parameters: type a number, then assign it with a SET key. All parameters survive restarts.

Compound interest with savings rate

SET K0 – stores the display as starting capital. Example: type 1000 → SET K0.

SET P% – stores the annual interest rate in percent. Example: 3 → SET P%.

SET JAHRE – stores the duration in years. Example: 10 → SET JAHRE.

SET RATE/M – stores the monthly savings amount. Example: 50 → SET RATE/M.

INFO – shows the summary: end value, total paid in and interest earned. Example (defaults): END 8,336 · IN 7,000 · INT 1,336.

ENDWERT – computes the final capital with monthly compounding: K0·(1+i)^m + rate·((1+i)^m−1)/i with i = P/1200 and m = years·12. Example: K0 = 1,000, P = 3, years = 10, rate = 50 → 8,336.42. The result goes to the history.

ZINSEN – shows only the interest earned (end value − K0 − all payments). In the example: 1,336.42.

RESET – restores all FIN parameters to defaults (K0 1000, P 3, years 10, rate 50, persons 2, rate 1.08).

Tip & bill splitting

TIP+10% – adds a 10 % tip. Example: 85 → 93.50.

TIP+15% – adds a 15 % tip. Example: 85 → 97.75.

TIP+20% – adds a 20 % tip. Example: 85 → 102.

SET PERS – stores the displayed number as the number of people (minimum 1, rounded). Example: 4 → SET PERS.

÷ PERS – divides the display by the stored number of people. Complete flow: 4 SET PERS → 85 TIP+15% → ÷ PERS → 24.44 per person.

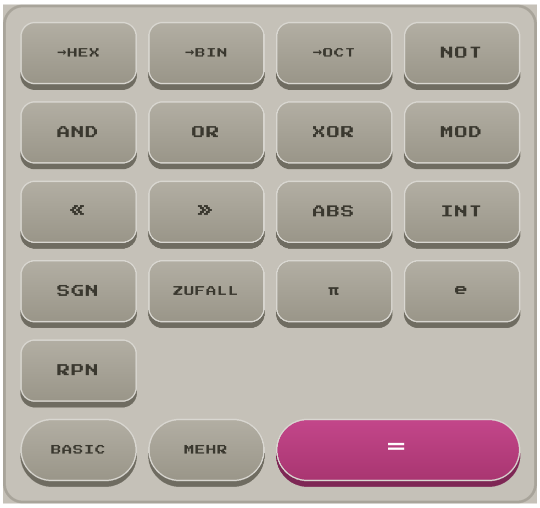
Currency

SET KURS – stores the display as the exchange rate (e.g. 1.08 for EUR→USD).

€→$ – multiplies by the rate. Example (rate 1.08): 100 → 108.

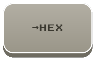
$→€ – divides by the rate. Example (rate 1.08): 108 → 100. Deliberately without live rates (nothing leaves the device); the rate works for any currency pair.

9. PRG PAGE (PROGRAMMER & RPN)



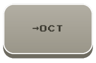
All bit operations use the integer part of the values (decimals are truncated).

Base display



→HEX – shows the integer part in hexadecimal in the status line; the display stays decimal. Example: 255 → HEX: FF.

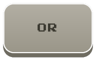
→BIN – display in binary. Example: 10 → BIN: 1010.



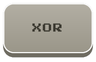
→OCT – display in octal. Example: 64 → OCT: 100.

Binary operators (same flow as ÷/×: first number operator second number =)

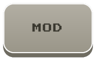
AND – bitwise AND. Example: 12 AND 10 = → 8 (1100 & 1010 = 1000).



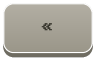
OR – bitwise OR. Example: 12 OR 10 = → 14 (1100 | 1010 = 1110).



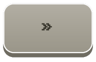
XOR – bitwise exclusive OR. Example: 12 XOR 10 = → 6 (1100 ^ 1010 = 0110).



MOD – remainder of a division. Example: 17 MOD 5 = → 2.



« – bit shift left (×2 per position). Example: 1 « 4 = → 16.



» – bit shift right (÷2 per position). Example: 255 » 4 = → 15.

Unary functions (apply immediately)

NOT – bitwise complement (~x). Example: 5 → NOT → −6.

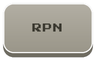
ABS – absolute value. Example: −7 → ABS → 7.

INT – truncates decimals. Example: 3.9 → INT → 3.

SGN – sign function: −1, 0 or 1. Example: −7 → SGN → −1.

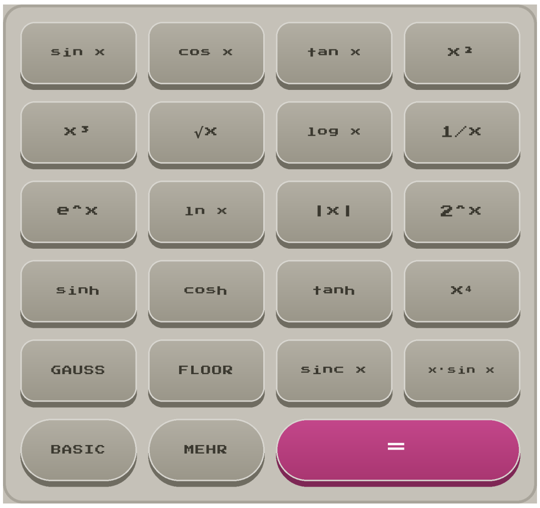
ZUFALL – random number between 1 and 100, e.g. for dice rolls. Example: ZUFALL → e.g. 37 (different every time).

RPN mode (Reverse Polish Notation)



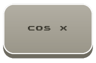
RPN – toggles the mode on/off (saved; the status line shows stack depth and top value). While active, the mode applies on every page. In RPN mode = pushes the displayed number onto the stack (max. 8 entries); an operator takes the top stack value as the first operand and combines it with the display: 3 = 4 + → 7. AC additionally clears the stack. Calculations appear in the history with an RPN prefix.

10. PLOT PAGE (FUNCTION GRAPHS)



Tap a function key – the graph is drawn as pixel art on the LCD, with axes (where visible) and automatic scaling. Any key or a tap on the display closes the plot. The 20 functions and their ranges:

sin x – sine wave, oscillates between −1 and 1; range −6.5…6.5.

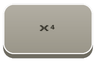


cos x – cosine, like sine shifted by 90°, starts at 1; range −6.5…6.5.

tan x – tangent with poles every 180° (blanked out); range −6.5…6.5.

x^2 – parabola with its vertex at the origin; range −4…4.

x^3 – cubic curve through the origin, point-symmetric; range −3…3.



x^4 – like x^2, but flatter at the origin and steeper outside; range −2.5…2.5.

√x – root curve, only defined for $x \geq 0$; range 0…16.

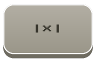
log x – base-10 logarithm, crosses the x-axis at 1; range 0.05…20.

ln x – natural logarithm, same shape, steeper; range 0.05…12.

1/x – hyperbola with a pole at 0 (blanked out); range −4…4.

e^x – exponential, crosses the y-axis at 1; range −3…3.

2^x – doubling curve, flatter than e^x; range −3…4.

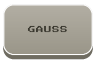


|x| – absolute value, V shape with a kink at the origin; range −4…4.

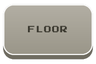
sinh – hyperbolic sine, S-shaped through the origin; range −3…3.

cosh – hyperbolic cosine, the catenary, minimum at 1; range −3…3.

tanh – hyperbolic tangent, S curve between −1 and 1; range −4…4.



GAUSS – bell curve $e^{(-x^2)}$, maximum 1 at $x = 0$; range −3.5…3.5.

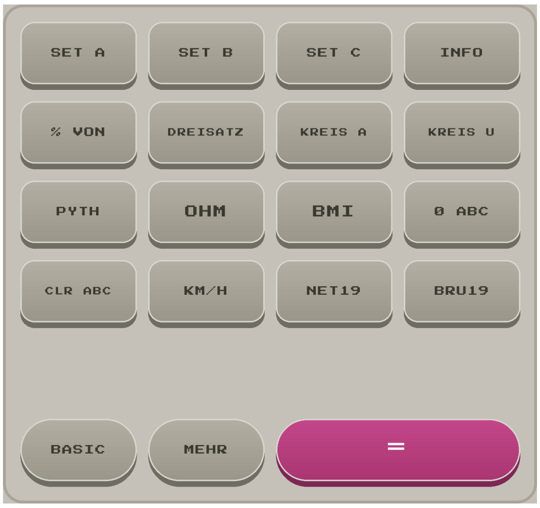


FLOOR – step function $\lfloor x \rfloor$, rounds down to the next integer; range −4…4.

sinc x – sin x / x, decaying wave with maximum 1 at the origin; range −15…15.

x·sin x – oscillation with outward-growing amplitude; range −12…12.

11. FORM PAGE (FORMULA ASSISTANT)



This page works with three variables A, B and C: type values and assign them, then press a formula key. The variables stay stored until overwritten or cleared. Every formula result goes to the history; the status line briefly shows the variable roles used.

Managing variables

SET A – stores the displayed number as variable A. Example: type 15 → SET A.

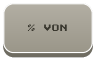
SET B – stores the displayed number as variable B. Example: 200 → SET B.

SET C – stores the displayed number as variable C. Example: 5 → SET C.

INFO – shows the current values of A, B and C. Example: A=15 B=200 C=0.

CLR ABC – resets all three variables to 0. Example: after CLR ABC, INFO shows A=0 B=0 C=0.

Formulas

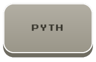


% VON – A percent of B: $B \cdot A/100$. Example: A = 15, B = 200 → 30.

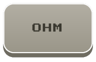
DREISATZ – rule of three A : B = C : x, result $x = B \cdot C/A$. Example: 3 kg cost 6 €, what do 5 kg cost? → A = 3, B = 6, C = 5 → 10.

KREIS A – circle area with radius A: $\pi \cdot A^2$. Example: A = 2 → 12.566.

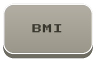
KREIS U – circle circumference with radius A: $2 \cdot \pi \cdot A$. Example: A = 2 → 12.566.



PYTH – hypotenuse from legs A and B: $\sqrt{(A^2+B^2)}$. Example: A = 3, B = 4 → 5.

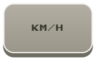


OHM – voltage from Ohm's law $U = R \cdot I$ with A = resistance (Ω) and B = current (A). Example: A = 100, B = 0.5 → 50 V.



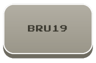
BMI – body mass index A / B^2 with A = weight (kg) and B = height (m). Example: A = 80, B = 1.8 → 24.69.

Ø ABC – average of the three variables: $(A+B+C)/3$. Example: A = 10, B = 20, C = 30 → 20.



KM/H – speed A / B with A = distance (km) and B = time (h). Example: A = 150, B = 1.5 → 100 km/h.

NET19 – converts gross→net (19 %): A / 1.19 with A = gross amount. Example: A = 119 → 100.



BRU19 – converts net→gross (19 %): $A \cdot 1.19$ with A = net amount. Example: A = 100 → 119.

12. GAMES

GAME

GAME opens the menu: 1, 2 or 3 starts MATH ATTACK at that level, 5 starts SNAKE, AC cancels.

MATH ATTACK

30 seconds of mental arithmetic. The task appears in the status line, type the answer, press =. Correct = 1 point; wrong counts as a mistake; either way the next task appears immediately. The status line shows remaining time (T) and score (S).

• Level 1 EASY: only + and − with small numbers (up to 20).
• Level 2 NORMAL: + − up to 99, × ÷ within the 12× tables.
• Level 3 HARD: + − up to 999, × ÷ up to 19.
• Each level keeps its own high score. A round without a single mistake ends with PERFECT! and its own jingle. AC quits early.

SNAKE

The classic on the LCD. Controls: 8 = up, 2 = down, 4 = left, 6 = right (or arrow keys). Eating food grows the snake, and the speed increases with every bite. Walls or your own tail end the game; the high score is saved. AC or GAME quits.

13. LANDSCAPE & KEYBOARD

Rotate the iPhone: the layout switches to display on the left, keypad on the right, and an additional quick-access column with sin, cos, tan, log, ln, x^2, 1/x, x^y, π and e appears next to the BASIC keypad. All extra pages work in landscape too.

On Mac/iPad with a keyboard: digits, + − * /, Enter = result, Esc = AC, % = percent, R or W = square root. Arrow keys steer SNAKE.

14. SECRET THEME UNLOCK

Enter:

8 8 2 2 4 6 4 6 +

This unlocks the Virtual Boy theme, stores it permanently in local storage and makes it available through the THEME button.

15. STORAGE & PRIVACY

Everything is stored locally in the browser only: theme, sound, angle mode, calculation history, high scores (per level and for SNAKE), memory value (M), FIN parameters, exchange rate, number of people, RPN mode with its stack, and the formula variables A/B/C. Nothing is transmitted anywhere; there are no external connections. To wipe everything, delete the website data for this domain in your browser.

16. TROUBLESHOOTING

No sound on the very first start: browsers block audio before the first touch – press any key once.

Old version after an update: the service worker caches the app; close the app/tab completely and reopen it twice.

Battery LED always red: the Battery API only exists in Chrome/Android. On iPhone the LED stays classic red – like the original Game Boy.

= stores numbers instead of calculating: RPN mode is active – turn it off with RPN on the PRG page.